# Controller RTL Verification

## VCS + Verdi | Original RTL, high-contrast stress fixture

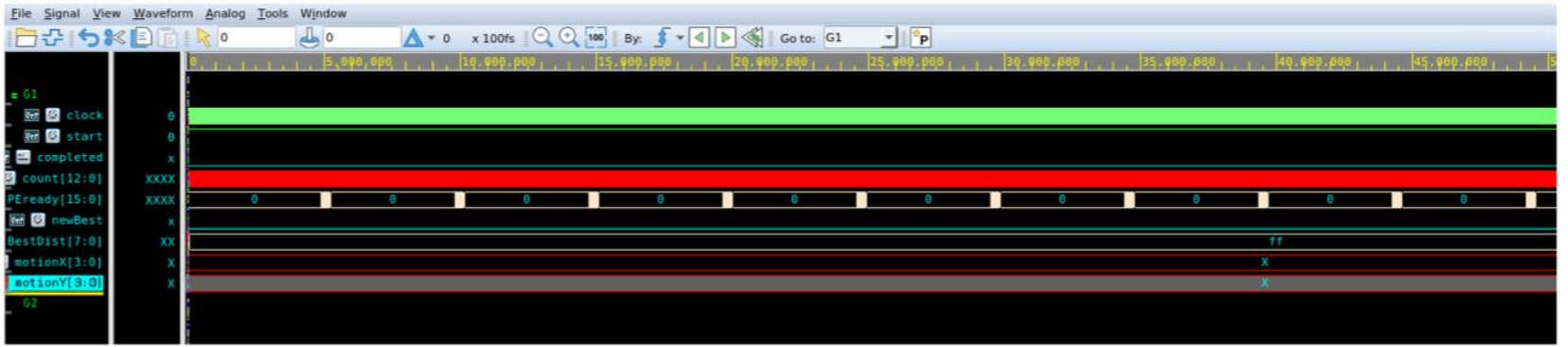
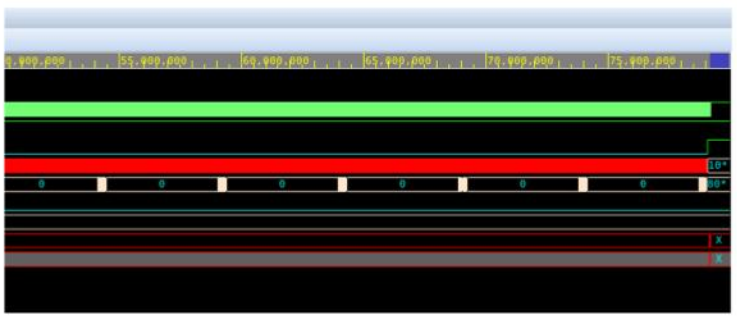



**Figure 1. Clock, PEready, count, newBest and output registers across the test.**

**Figure 2. End of the simulation window**



### Original 8-bit result

**BestDist = FF**

**motionX, motionY = X, X**

The saturated comparator never chooses a valid candidate in this stress test.

# 16-bit SAD Verification

## VCS + Verdi | Separate precision-corrected RTL and directed stimulus

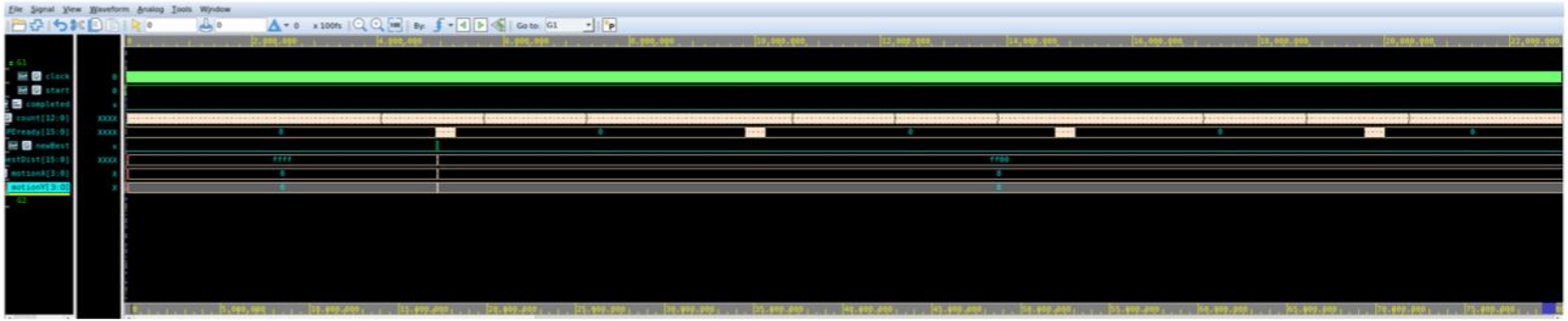


Figure 3. newBest selects FF00 and the encoded vector (8,8) for the all-FF stress case.

**BestDist is input-dependent: these are three different stimulus cases.**

| | |
|---|---|
| Team perfect match | **0000** |
| Single-pixel perturbation | **0003** |
| All-00 reference / all-FF search | **FF00** |

**FF00 is not the expected output of the team's 0000 perfect-match testbench.**

# 14 nm Physical Design Checkpoint

**Fusion Compiler + PrimeTime | 520 MHz | Routed V5 results**

## Physical implementation

| Measured metric | V5 result |
|---|---|
| Routed cell area | 2,781.30 µm² |
| Mapped cells | 6,310 |
| Sequential cells | 449 |
| Memory macros | 0 |
| Typical power | 0.729 mW |

## PrimeTime: extracted RC

| Corner | Setup WNS | Hold WNS |
|---|---|---|
| FAST | +0.52 ns | -0.14 ns |
| TYPICAL | +0.30 ns | -0.12 ns |
| SLOW | -0.06 ns | -0.09 ns |

**Setup: 17 SLOW failures**
**Hold: 402 FAST / 120 TYP / 33 SLOW**

**V5: CTS and routing completed. ICV DRC still reports 18 violations.**

Final sign-off requires DRC repair, physical ECO, extraction and PrimeTime setup/hold on the corrected layout.